Fast Fourier Transform

16 Point VLSI FFT

Project Description

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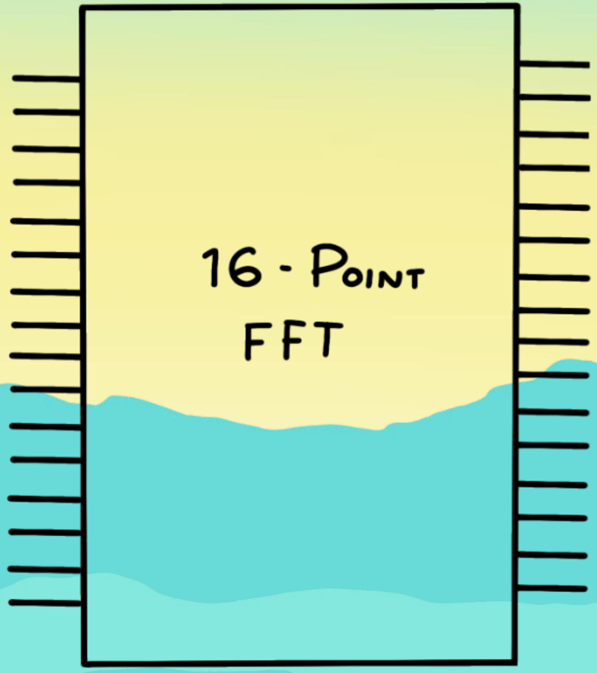
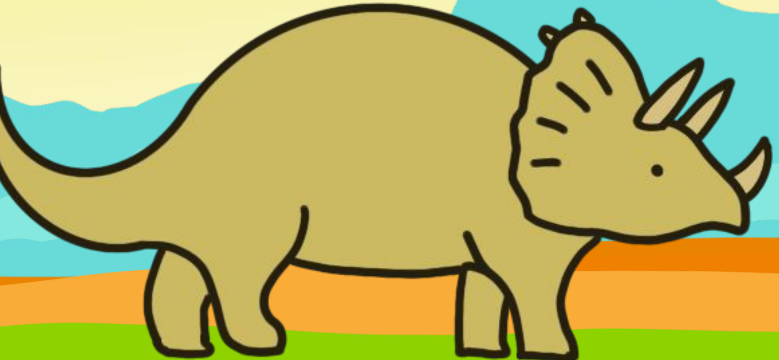
We designed and implemented a parallel 16-point Fast Fourier Transform (FFT) "processor" using SkyWater CMOS standard cells.

The processor is completely asynchronous and calculates everything with 8 bit precision meaning the total amplitude represented is 40db.

The input can also take both real and imaginary components in case this cell would like to be used in a larger system.

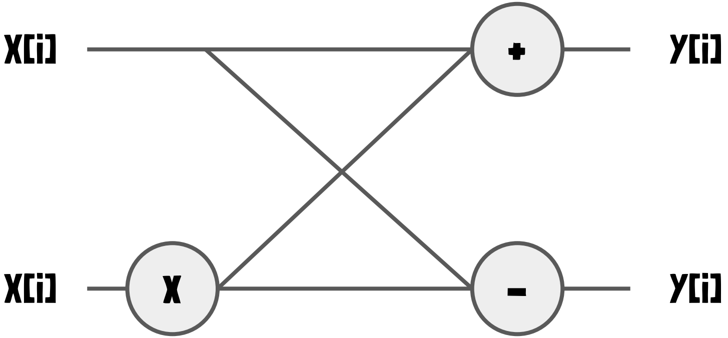


Algorithm Design and Approach



Architecture and Background

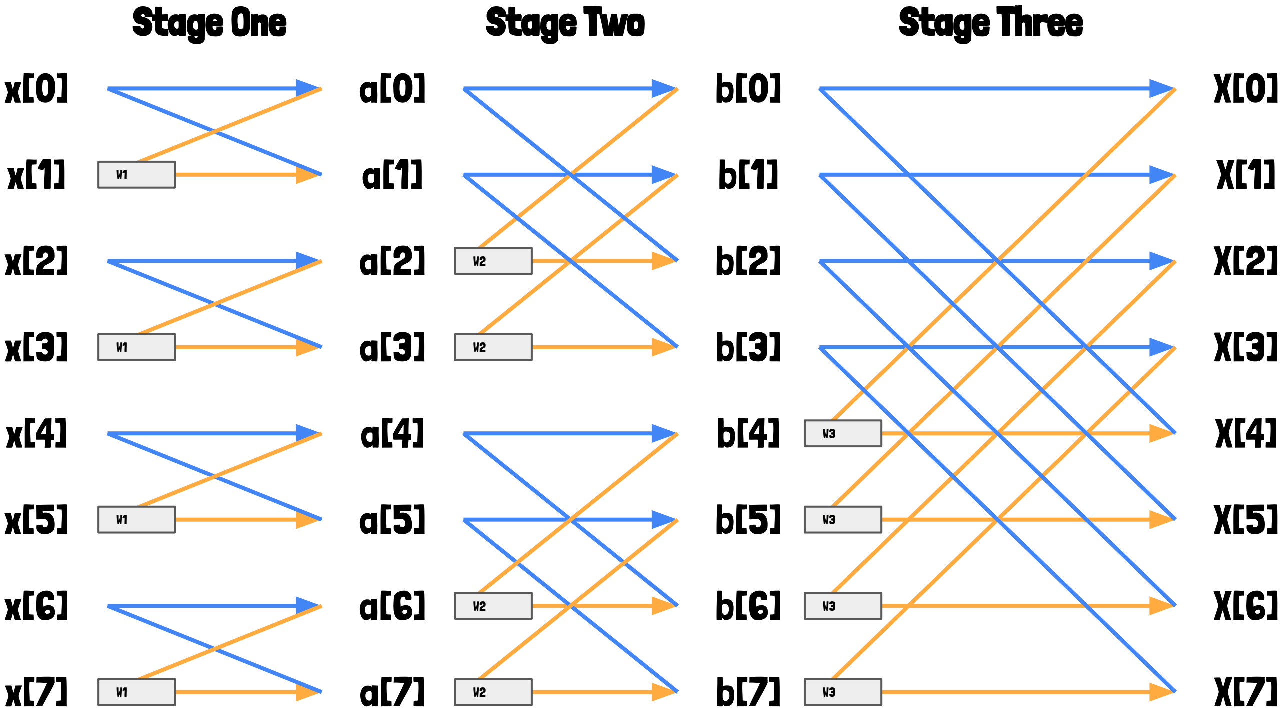
The FFT algorithm uses a recursive structure referred to as a "butterfly". The butterfly operates on two points as show. Where X[i] is a selected point and W[0] is a "twiddle" factor:



Using fixed point, the butterfly unit uses four multipliers to implement a unit that performed complex number multiplication, and regular adders to perform the appropriate additions. In order to decrease complexity, all real and imaginary components were implemented with 8 bit precision, leading to a maximum amplitude representation of 40db. Internally, each complex number is split into its real and imaginary components and kept separate for all calculations.

Applying Recursion

An Eight Point Example

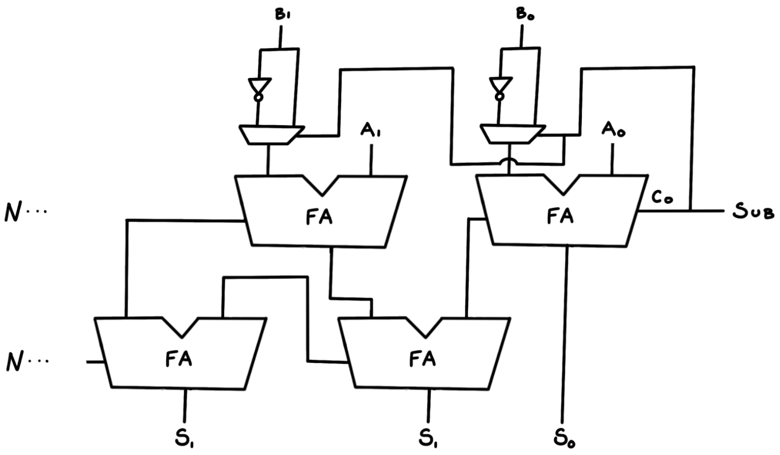


Circuit Design

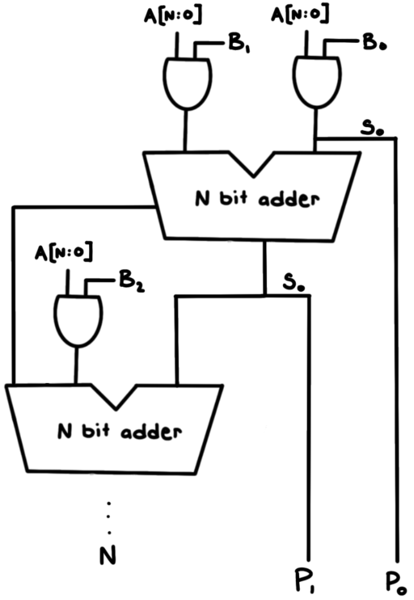
Basic Blocks

Adder And Multiplier

Carry Save Adder/Subtractor



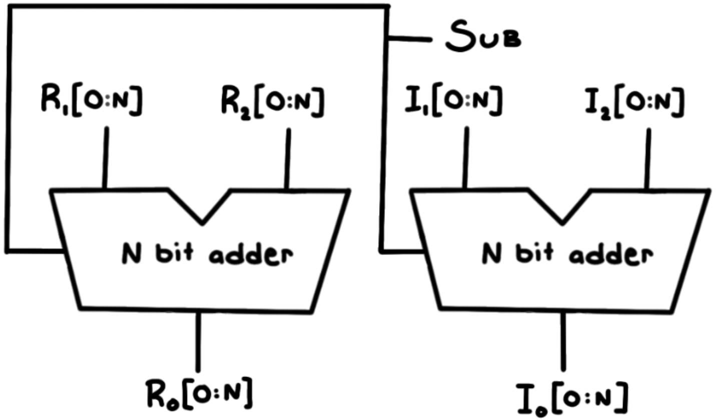
Multiplier



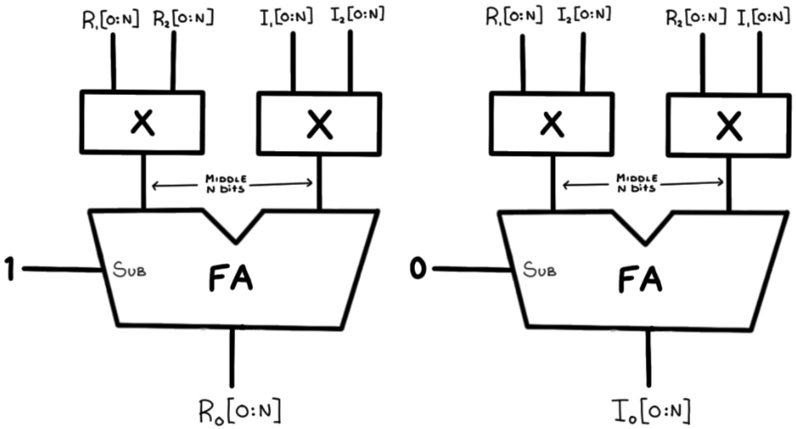
Component Blocks

Complex Number Calculations
R1+I1 (i) and R2+I2(i)

Complex Number Adder

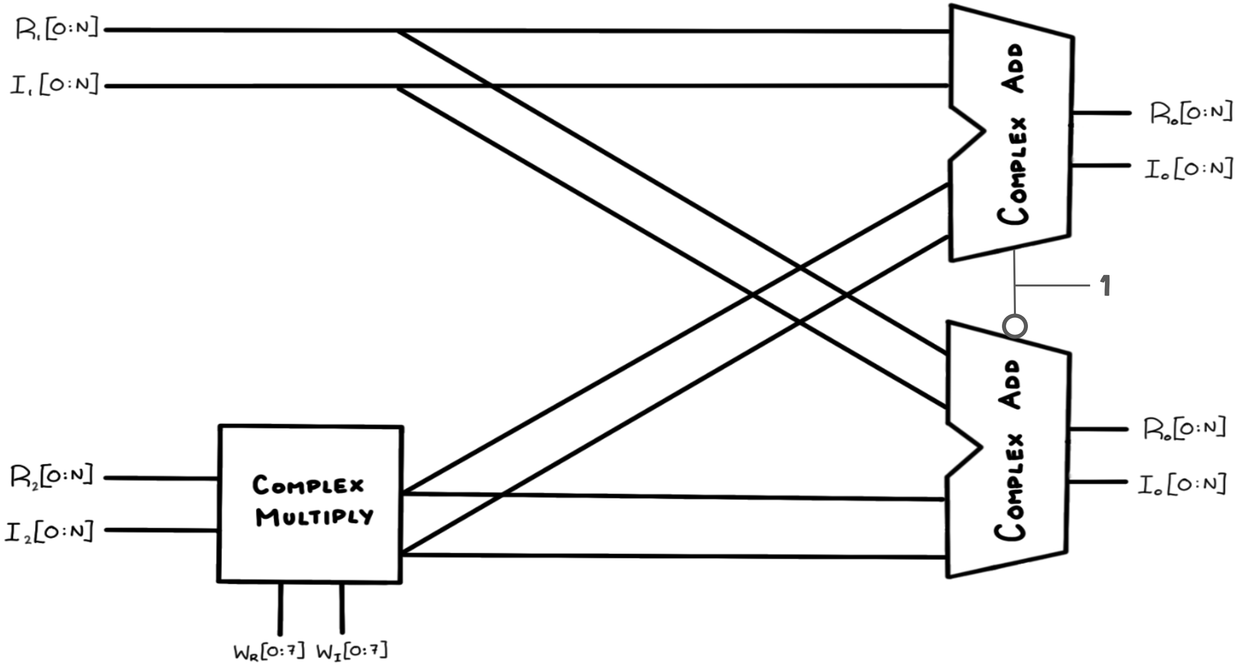


Complex Number Multiplier



Higher Level Components

Butterfly Cell

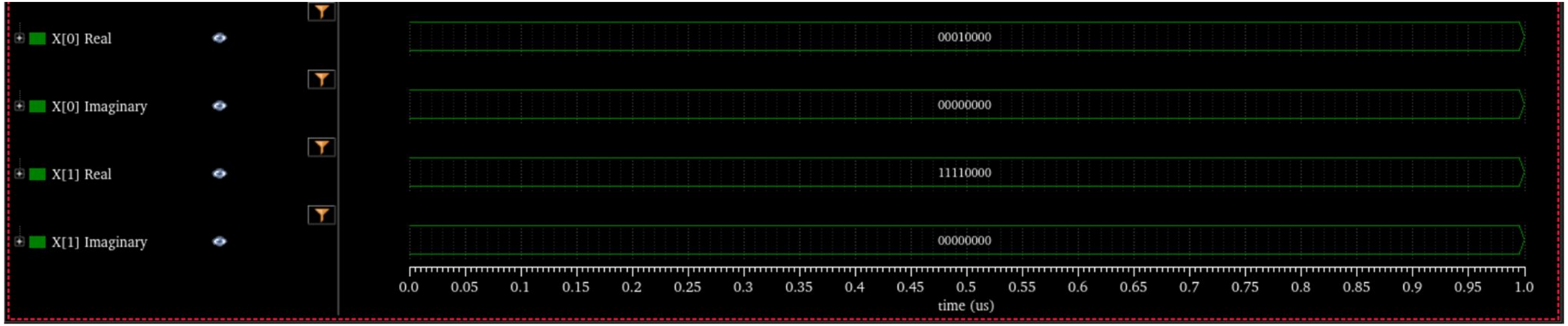


Simulation Results and Benchmarks

Simulating the Butterfly Cell

2 point FFT Sample Output

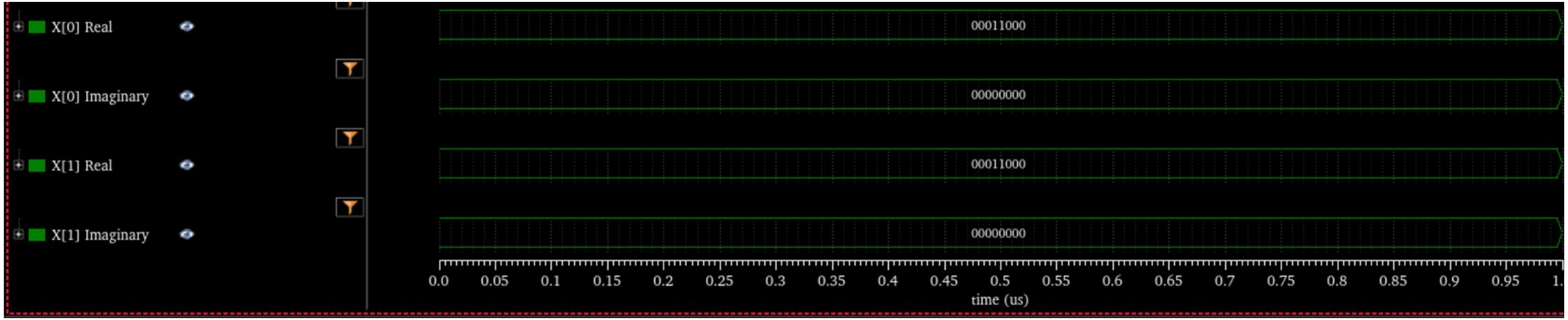
Sample Input to Output Mapping
[0,1] to [(1+0j),(-1+0j)]



Simulating the Butterfly Cell

2 point FFT Sample Output

Sample Input to Output Mapping
[1.5,0] to [(1.5+0j),(1.5+0j)]

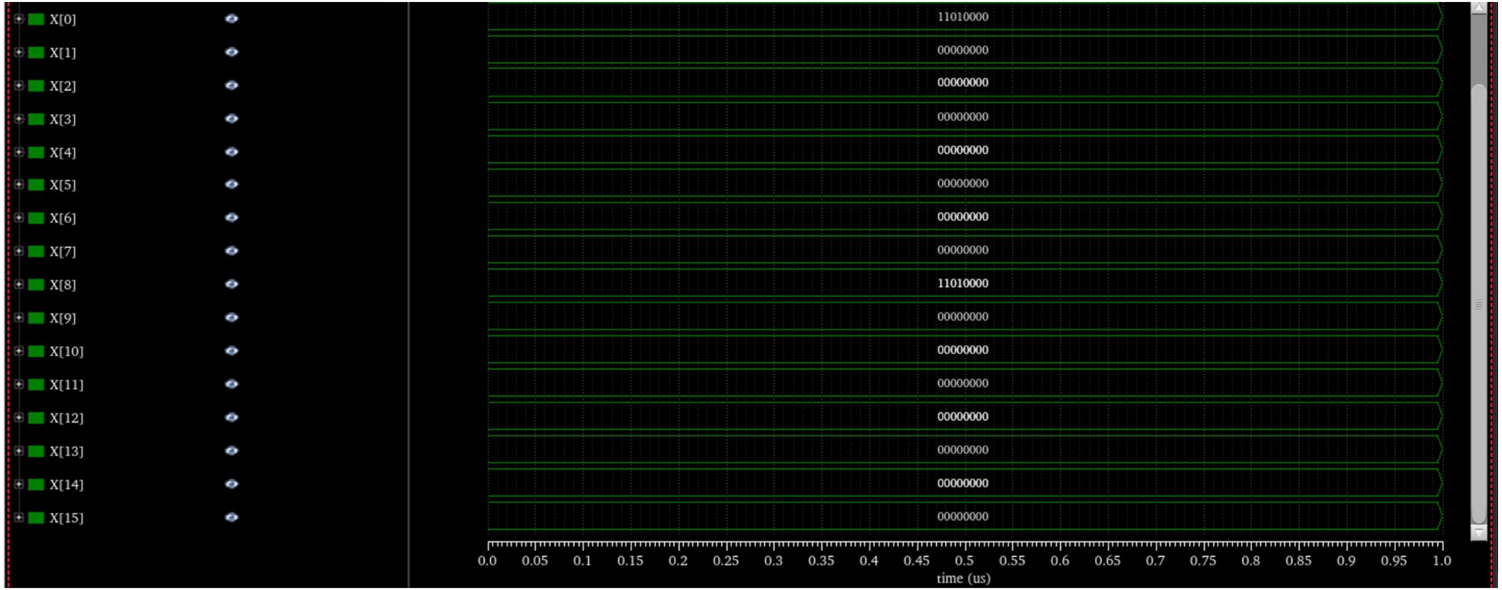


Simulating the Basic Cell

16 point FFT Sample Output

Sample Input to Output Mapping

[-0.375+0.j , 0.+0.j]x8 to [-3,0,0,0,0,0,0,0]x2



Circuit Delay (Asynchronous)

Butterfly

Full adder: ~47ps

CSA: ~412ps

Multiplier: ~3542ps

Total Delay: ~5190ps

16 Point

Conservative Estimate

Stage Delay: ~5.19ns

Total Delay:~21.99ns

Lessons Learned

Lessons

Circuit Design

- Delay builds up fast
  - Saw how much difference fast hardware can make in delay propagation
- Synchronous circuits might allow you to clock this circuit and process more information at the same time

Design Approach

- Smaller blocks is better
  - Verifying smaller schematics from the ground up is useful for debugging
- Specifications are important
  - Without setting clear boundaries for ability of the circuit, designing can be hard

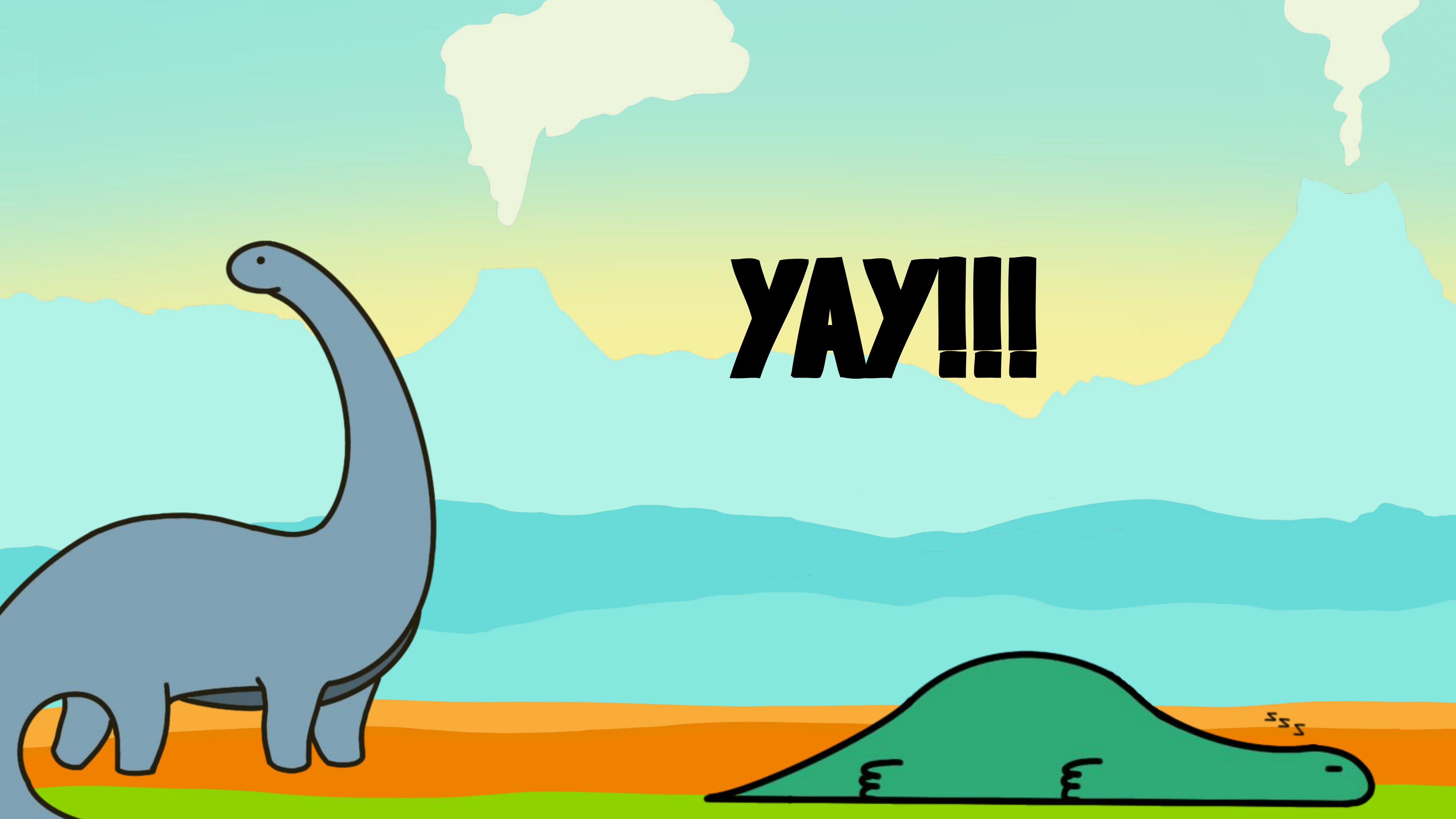
Future Improvements

Testbench

- Better testing
  - Utilizing ocean scripts
  - Clearer input/output comparisons
  - Automating testing
- Testbench for input variance
  - Vary input and test for circuit's resilience to input fluctuation

Circuit Improvements

- Since we had no area requirements and we were only using 8 bits, could have improved delay through a more complex, delay-optimized adder.
- Since we do not use the whole multiplier could have saved space and built a more use-case optimized multiplier.


YAY!!!

VIDEO LINK

https://youtu.be/lJYp3PPFcsc